

	Brown planthopper	Nilaparvata lugens
	White backed plant hopper	Sogatella furcifera
Important Pests of Vegetable crops		
Brinjal: Brinjal shoot and fruit borer	Lucinodes orbonalis	
Potato: Potato Tuber Moth	Phthorimaea operculella	
Sweet potato: Sweet potato weevil	Cylas formicarius	
Tomato: Tomato fruit borer	Helicoverpa armigera	
Okra: Spotted bollworms	Earias insulana, E. vitelli	
Red cotton bug	Dysdercus koenigii	
Cabbage and Cauliflower		
Diamond Back Moth	Plutella Xylostella	
Cabbage butterfly	Pieris brassicae	
Gourds		
Red pumpkin beetle	Aulacophora foveicollis	
Melon fruit fly	Bacterocera cucurbitae	
Onion		
Onion thrips	Thrips tabaci	
Onion maggot	Delia antiqua	

❖ **Polyphagous pests:** Locust, grasshopper, termites, hairy caterpillars, cutworms, thrips

1. **Locust:** Patanga succincta (Bombay locust)

- Locust migratoria (Migratory locust)
- Schistocera gregaria (Desert locust)

2. **Grasshopper**

- Oxya nitidula (Sporadic occurrence)

DEFENCE MECHANISMS OF INSECTS

I. Behavioural:

- (i) **Thanatosis** - insects pretends as if dead. e.g., some beetles
- (ii) **Structural** e.g., hardened forewings of beetles known as elytra protect the beetles from predation of birds.
- (iii) **Colourational:** Presence of protective colours. e.g., Stick insects
- (iv) **Chemical:** Presence of defensive chemicals. e.g., Bees producing venom

❖ **The following are some examples of waiting period of some chemicals in a few important crops:**

Crop	Insecticide	Dose	Waiting period (days)
Chillies	Dicofol	0.05%	1
	Quinalphos	0.05%	8
Tomato	Phosalone	0.05%	3
	Quinalphos	0.05%	5
Brinjal	Phosalone	0.05%	2
	Endosulfan	0.07%	3
	Aldicarb	1 kg a.i./ha	60